

FRA Atlas and WebGIS-based DSS

Good morning, esteemed judges and fellow innovators. I am Aryan, representing Team EvoNex. Today, I am excited to present “**FRA Atlas and WebGIS-based Decision Support System (DSS)**”, the problem statement for our team. Our AI-powered solution for transforming the Forest Rights Act (FRA) implementation.

The Challenge: Fragmented FRA Data

The Forest Rights Act of 2006 was a **landmark law** that recognized the land rights of tribal and traditional forest communities. It empowers Gram Sabhas (village councils) to claim and protect ancestral lands. In theory, FRA decentralizes and democratizes forest governance. In practice, however, the system is overwhelmed by **scattered, non-digitized records** and the absence of a unified FRA Atlas. As of January 2025, nearly *36% of the 50 lakh FRA claims* have already been rejected, and about *7.5 lakh applications remain pending*. Manual processes and legacy paper archives are to blame: claims get delayed for years, transparency is low, and forest communities wait painfully long for justice.

These challenges slow down land titling, sow confusion, and can even lead to unlawful evictions. With new government schemes pushing for **digitization and mapping** of FRA rights, , the need for a centralized, tech-driven solution has never been more urgent.

Proposed Solution: The EvoNex Platform

To tackle this, **EvoNex** has developed a single, integrated platform that brings together all FRA data and tools. In our vision, a user-friendly WebGIS interface will display *live* information: verification status, claim maps, and updates in real time. Every piece of information about a claim – from the original filed form to current sanction status – is retained in a **persistent claim history**, ensuring that officials and claimants always have full context.

Our system ingests legacy paper records through AI-powered OCR and NLP, geocodes the claimed parcels, and overlays them on map layers. Satellite imagery and historical land records flow in seamlessly. In effect, EvoNex provides a **centralized FRA Atlas** that literally paints a clear picture of every claim in space and time.

Key Features and Unique Selling Points

EvoNex stands out for its innovative integration of disparate data and smart verification. Our **unique selling points** include:

- **Data Orchestration:** We unify legacy FRA records, GIS layers (forest parcels, village boundaries, etc.), and satellite data into a single workflow. No more silos – all relevant information travels together.
- **Cross-System Integration:** EvoNex connects government databases (land records, beneficiary lists, remote sensing feeds and related schemes) with our DSS. This means that when a claim is processed, it automatically pulls in the latest survey maps and scheme eligibility data.
- **Built-in Verification Suite:** Automated checks are performed at every step – for example, flagging overlapping claims, ensuring boundaries align with official records, and validating land use data. This dramatically **accelerates processing** and improves accuracy.
- **Transparency & Accessibility:** The GIS dashboard makes claim statuses visible to all stakeholders – forest officials, tribal communities, and local administrators. This promotes trust and inclusive decision-making.
- **Scalability and Testing:** Our architecture is containerized and cloud-ready, enabling seamless system testing, updates, and future expansion as more states are onboarded.

In a pilot run, these features cut the average FRA claim processing time from **12 months to just 2 months**. In fact, leveraging GIS and AI is known to *automate data collection and reduce processing time*. By speeding up verifications and reducing human error, EvoNex not only enforces FRA faster but also ensures that rightful claimants are served without delay.

Feasibility Analysis & Viability

Our feasibility study shows EvoNex is ready for implementation:

- **Low Running Costs:** We estimate monthly cloud costs of only \$200–\$500 for API usage, plus \$50–\$100 for hosting and microservices. Databases (PostgreSQL, MongoDB or Supabase) and GIS layers require minimal additional storage.
- **Technology Stack:** We rely on proven open-source tools: PostgreSQL/PostGIS and GeoServer for spatial data, React and Mapbox for the front end, and FastAPI or equivalent for backend services. These are industry-standard and well-supported.
- **Risk Mitigation:** We have strategies for each risk. For example, to control OCR and GIS API costs, we implement result caching and batch processing. Data accuracy is ensured via regex checks and cross-validation with government data. To handle scaling, our microservices architecture and load balancing allow the system to grow with demand.
- **Performance Targets:** Even with a small footprint (a containerized GIS instance under 1000GB memory), the system achieves response times under 1.5 seconds for initial queries and under 1 second thereafter. This ensures a smooth user experience.

With clear cost estimates and technical solutions mapped out, **EvoNex is both technically and economically viable**. In fact, our demo is already available and gives a taste of the live system.

Technical Approach

Let me walk you through the data flow. On the left, **Input** can be a scanned claim form or a query. Admins log in to manage data. Behind the scenes, our **processing pipeline** kicks in:

1. **Document Processing:** We use OCR (Optical Character Recognition) and NLP to extract claimant details from old paper forms or PDFs.
2. **Database & GIS Linking:** Extracted data is saved into our database. We then geocode the claimed coordinates and overlay them on GIS layers (forest parcels, village polygons, etc.).
3. **Verification Engine:** The system automatically checks for anomalies – e.g., overlapping claims, encroachments, or mismatches with satellite imagery. Suspect cases are flagged for review.

On the right, **Output** is a rich interactive Atlas. Officials can **plot tasks** on the map, see claim histories, and generate reports. Meanwhile, the system generates **insights and recommendations**: for instance, identifying regions with high conflict risk or spotting patterns that inform policy. By combining satellite data, socio-economic layers, and FRA records, EvoNex empowers data-driven decisions for better governance.

Every component – from mobile data collection to map visualization – is containerized and standardized. This ready use of cloud-native design means we achieve a high Technology Readiness Level.

Impact, Benefits & SDGs

The impact of EvoNex is transformative:

- **Transparency:** Claim statuses and decisions become clear and accessible to both officials and forest communities. By making data visible, we rebuild trust.
- **Inclusion:** Our pilot covers ~5,000 FRA claims across several districts. Real-time updates allow Gram Sabhas to see progress on their claims, empowering marginalized groups.
- **Efficiency:** In the pilot, *5,000 claims* were digitized and managed over 4 GIS layers (forest parcels, claim points, village boundaries). This digitization cut processing time from 12 months to just 2 months on average – a sixfold improvement. As geospatial experts note, GIS and satellite data can *automate data collection and reduce processing time*, which we have demonstrated.
- **Environmental and Social Governance:** By securing land rights faster and transparently, EvoNex directly supports **SDG 10** (Reduced Inequalities) by empowering tribal communities. It also aligns with **SDG 15** (Life on Land) since sustainable forest management relies on recognizing indigenous land rights sdgs.un.org. Moreover, providing open data tools and clear procedures promotes **SDG 16** (Peace, Justice and Strong Institutions) through accountable governance.

Additional benefits include proactive **dispute support**: GIS analysis alerts stakeholders to potential claim overlaps before conflicts arise. Communities gain **empowerment** through data – they no longer navigate opaque systems blindly. And the government benefits from **resource optimization** by targeting interventions where they are needed most, based on the insights our system provides.

Conclusion

In conclusion, the EvoNex FRA Atlas and WebGIS DSS represent a **paradigm shift** in implementing the Forest Rights Act. We leverage AI, cloud computing, and open

geospatial standards to solve the pressing problems of data fragmentation and processing delays. The result is a faster, fairer, more transparent system that honors the FRA's original vision of justice for forest dwellers.

We invite the Smart India Hackathon community to join us in bringing this solution to life. With a demonstrated pilot and solid architecture, we are ready to scale this up. Together, we can ensure that the FRA truly fulfills its promise of empowering communities and conserving our forests.

Thank you for your attention. I welcome any questions.

Research & References

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